Climate CHANGE per interval − four ponds through the Holocene, survey plots in the Anthropocene

Holocene = UNCORRECTED CHELSA−TraCE21k (the series the analysis uses), 500−yr rolling mean across the 5 moving−window offsets, which are drawn faint behind.
Right of the dotted line: Anthropocene change across survey plots (present − past), one box per site. All paces are PER CENTURY at both ends.

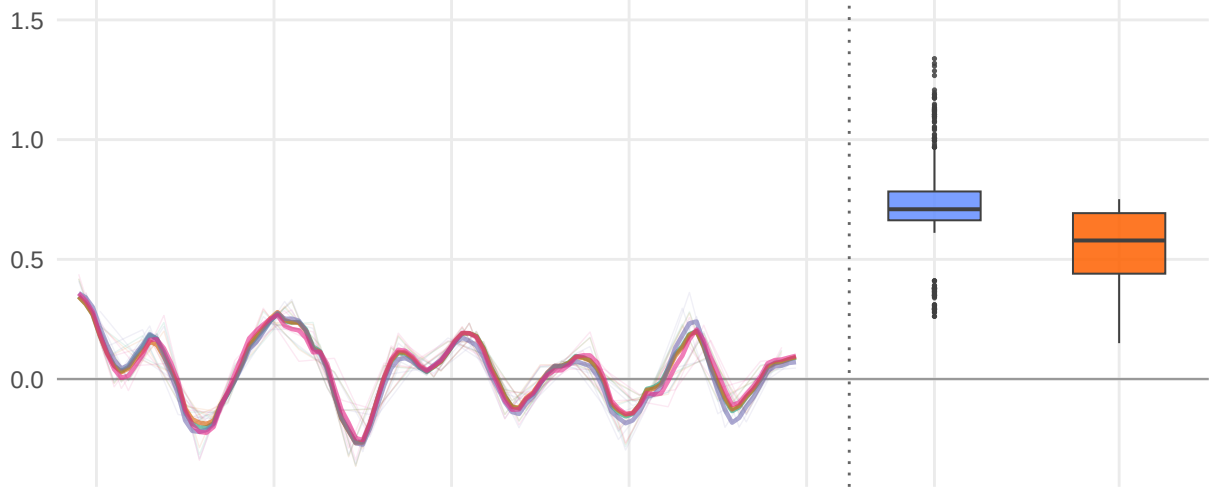
Site (Anthropocene): MORA, NOCA — Pond (Holocene): JayBath, LogWallow, PantherPotholes, ReflectionPond

MAT

degC per century

	min	mean	max
JayBath	−0.072	0.007	0.084
LogWallow	−0.073	0.007	0.084
PantherPotholes	−0.071	0.005	0.082
ReflectionPond	−0.068	0.007	0.087

MORA	0.952	2.474	4.249
NOCA	0.545	1.932	2.636

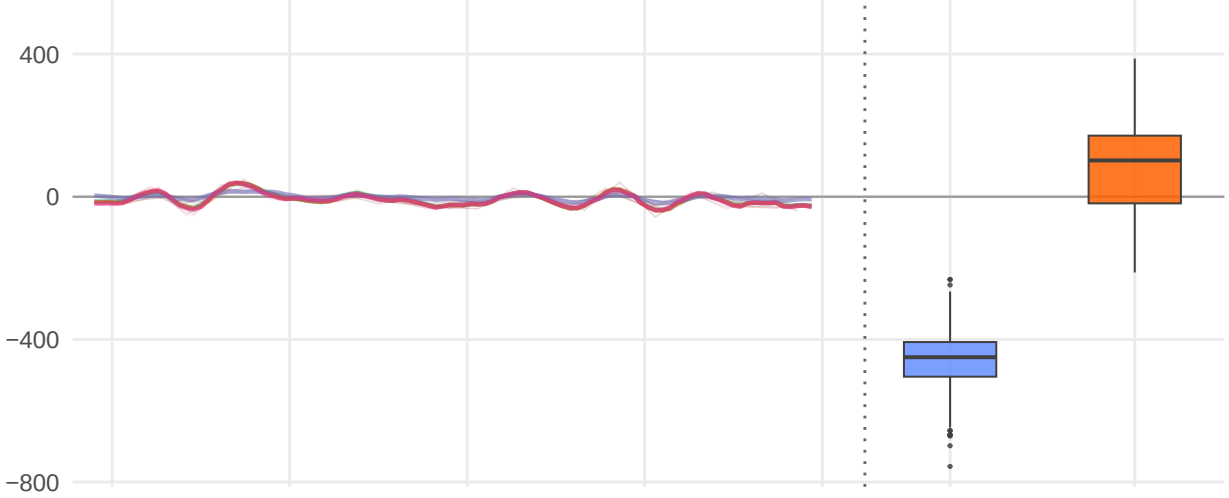


AP

mm per century

	min	mean	max
JayBath	−11.2	−1.513	9.680
LogWallow	−11.7	−1.637	10.0
PantherPotholes	−5.280	−0.323	4.000
ReflectionPond	−11.4	−1.669	9.560

MORA	−2401	−1570	−844
NOCA	−745	321	1359

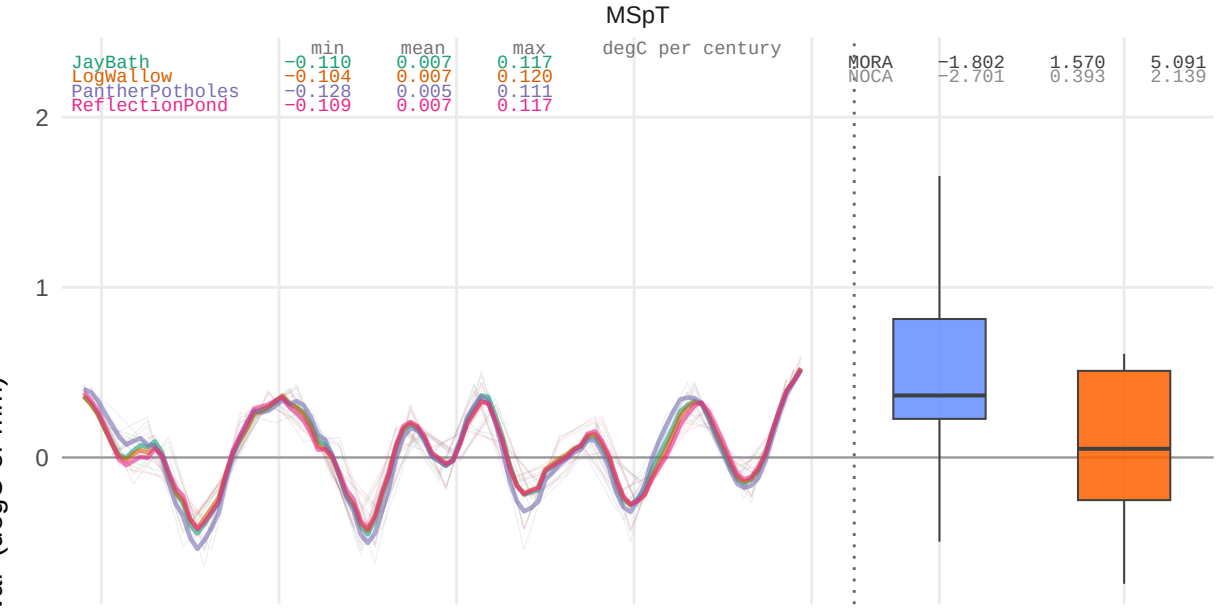


MSpT

degC per century

	min	mean	max
JayBath	−0.110	0.007	0.117
LogWallow	−0.104	0.007	0.120
PantherPotholes	−0.128	0.005	0.111
ReflectionPond	−0.109	0.007	0.117

MORA	−1.802	1.570	5.091
NOCA	−2.701	0.393	2.139

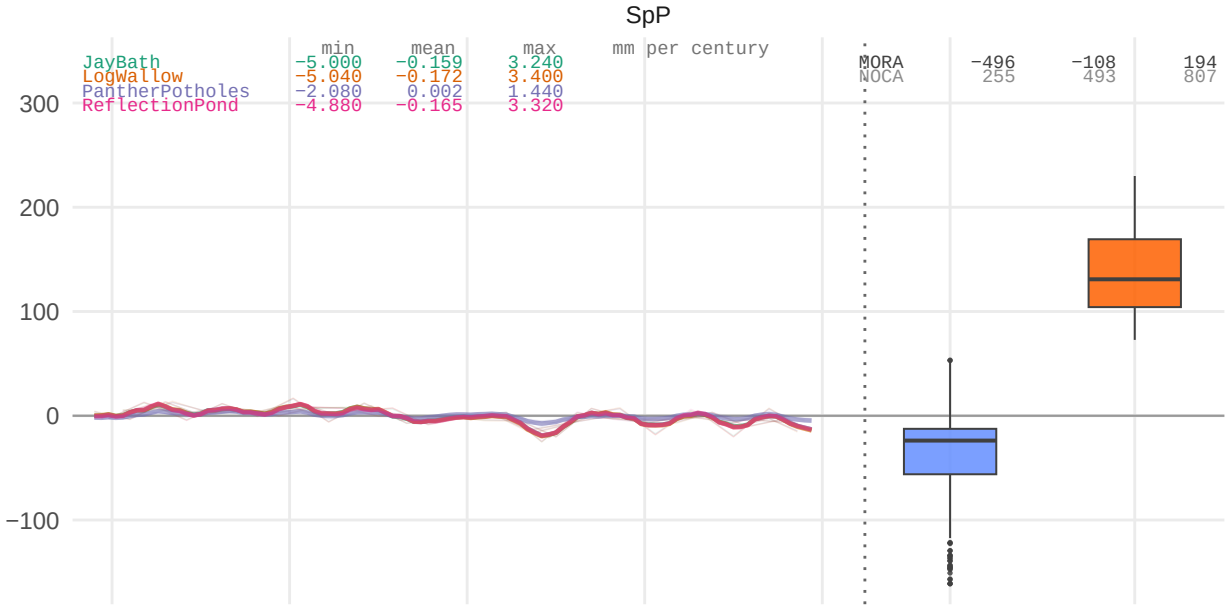


SpP

mm per century

	min	mean	max
JayBath	−5.000	−0.159	3.240
LogWallow	−5.040	−0.172	3.400
PantherPotholes	−2.080	0.002	1.440
ReflectionPond	−4.880	−0.165	3.320

MORA	−496	−108	194
NOCA	255	493	807



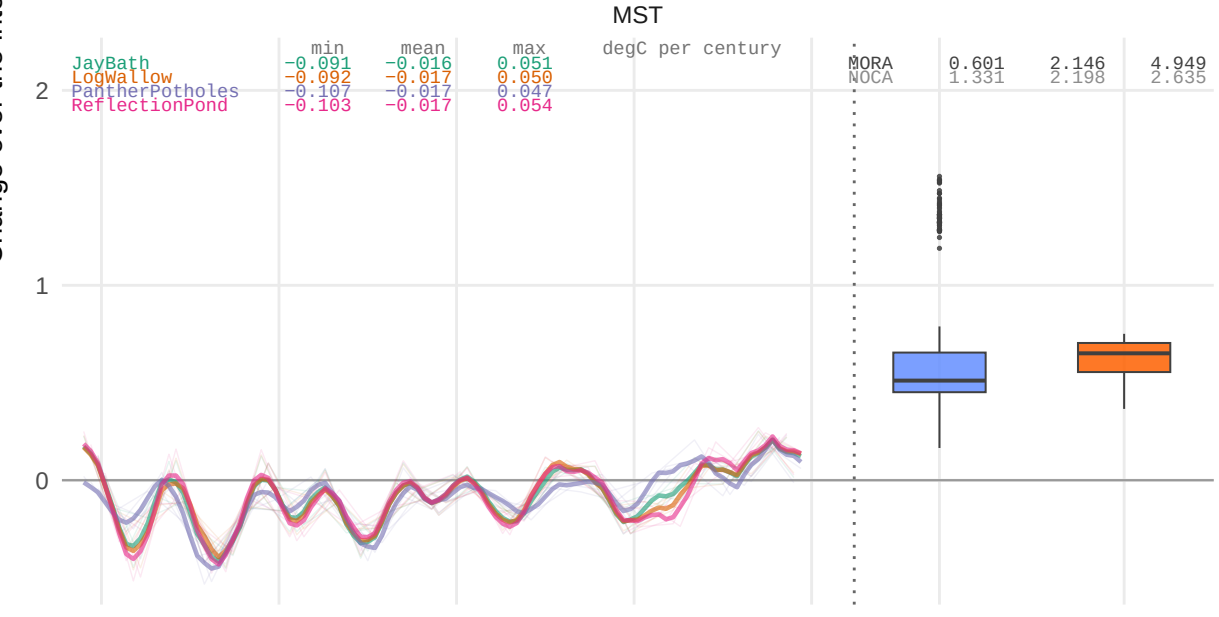
Change over the interval (degC or mm)

MST

degC per century

	min	mean	max
JayBath	−0.091	−0.016	0.051
LogWallow	−0.092	−0.017	0.050
PantherPotholes	−0.107	−0.017	0.047
ReflectionPond	−0.103	−0.017	0.054

MORA	0.601	2.146	4.949
NOCA	1.331	2.198	2.635

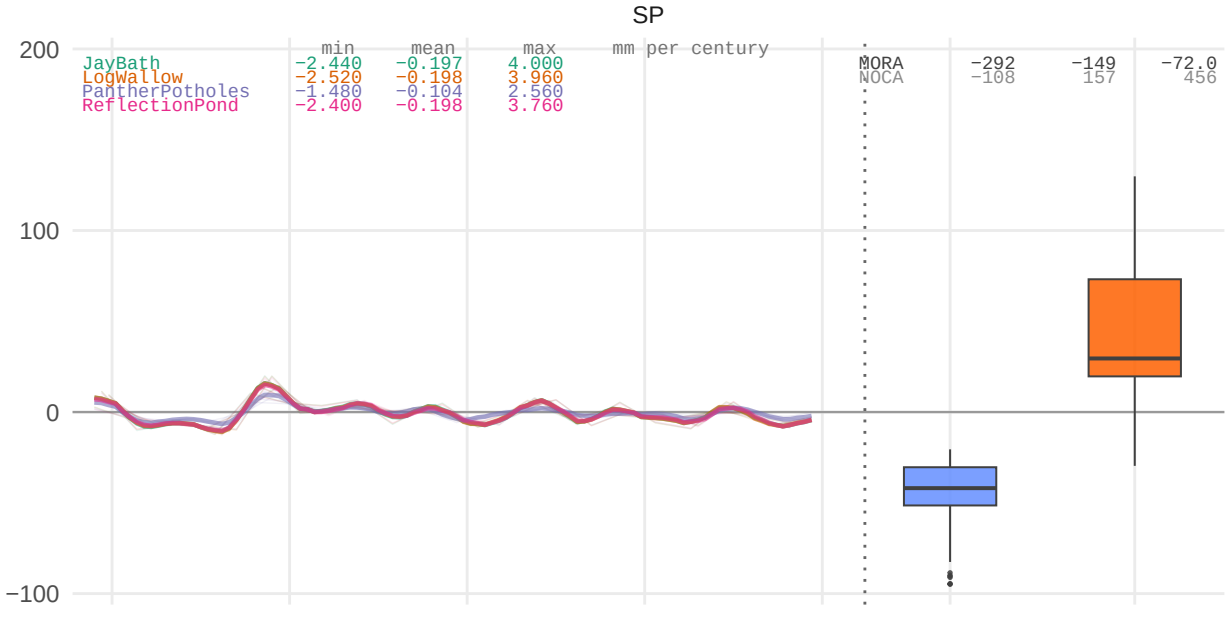


SP

mm per century

	min	mean	max
JayBath	−2.440	−0.197	4.000
LogWallow	−2.520	−0.198	3.960
PantherPotholes	−1.480	−0.104	2.560
ReflectionPond	−2.400	−0.198	3.760

MORA	−292	−149	−72.0
NOCA	−108	157	456

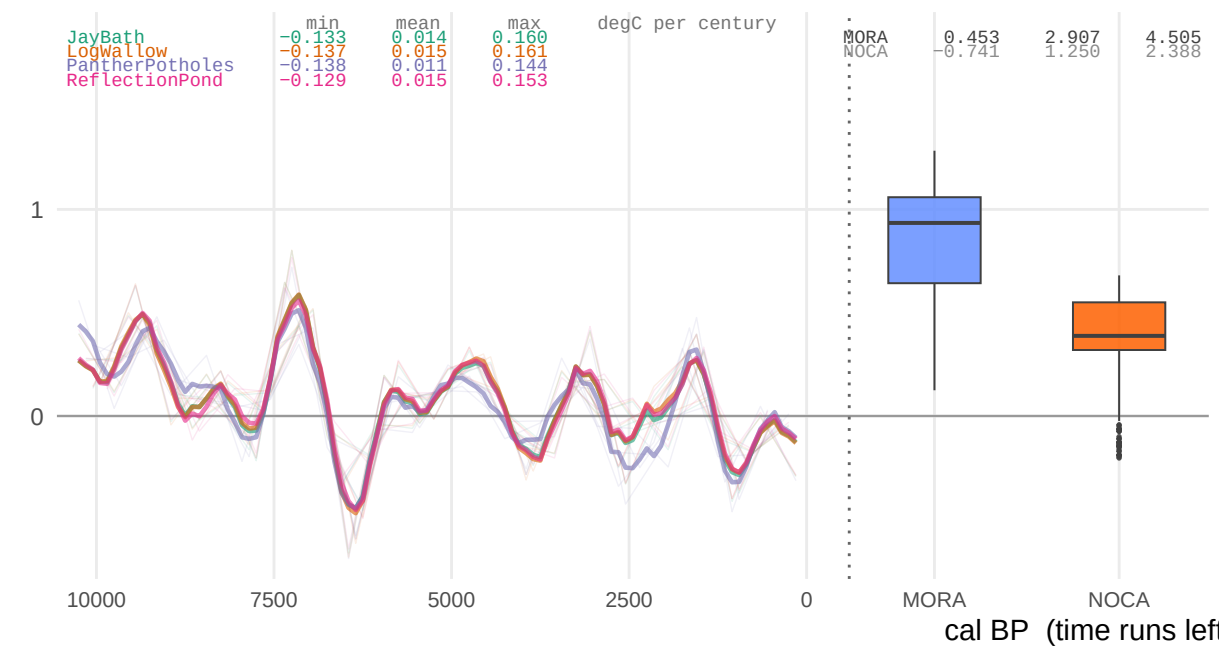


MWT

degC per century

	min	mean	max
JayBath	−0.133	0.014	0.160
LogWallow	−0.137	0.015	0.161
PantherPotholes	−0.138	0.011	0.144
ReflectionPond	−0.129	0.015	0.153

MORA	0.453	2.907	4.505
NOCA	−0.741	1.250	2.388

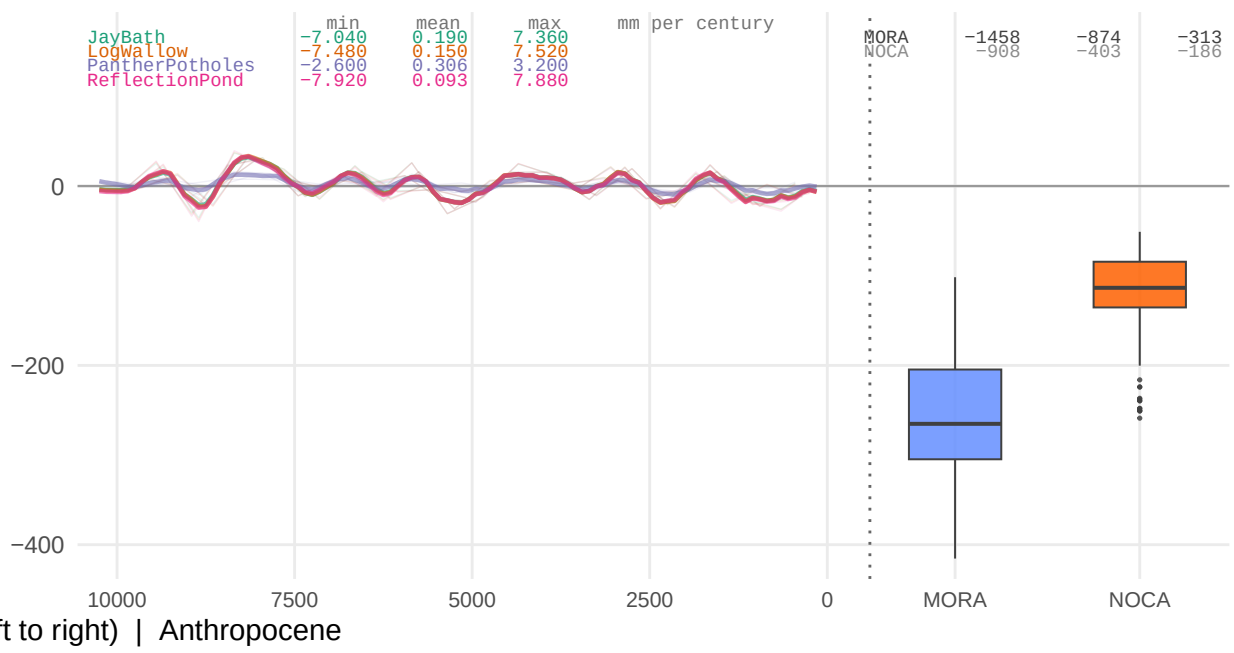


WP

mm per century

	min	mean	max
JayBath	−7.040	0.190	7.360
LogWallow	−7.480	0.150	7.520
PantherPotholes	−2.600	0.306	3.200
ReflectionPond	−7.920	0.093	7.880

MORA	−1458	−874	−313
NOCA	−908	−403	−186



cal BP (time runs left to right) | Anthropocene

Holocene = the FULL TraCE21k record at each pond (all 111 slices, ~100 yr apart), NOT the ages where cores happen to have samples − so the line is the location's climate history, independent of coring. Cores themselves reach JayBath 6841, LogWallow 6808, PantherPotholes 10609, ReflectionPond 5932 cal BP. Intervals differ (500 yr vs ~30 yr): these are CHANGES, not rates − see the

Correlation between climate variables, Holocene vs Anthropocene

The CCA learns the species−climate relationship from the ANTHROPOCENE plots and applies it to HOLOCENE samples.
Contrasting covariance structures can limit transfer; correlations alone do not test predictive validity.

Sign flips with |r| > 0.2 in both eras: MWT~MST (−0.58 vs 0.85); MST~MWT (−0.58 vs 0.85); AP~MWT (0.47 vs −0.21); SP~MWT (0.41 vs −0.28); WP~MWT (0.54 vs −0.21); MWT~AP (0.47 vs −0.21); MWT~SP (0.41 vs −0.28); MWT~WP (0.54 vs −0.21)